F·O·R·G·E

Functional-posterior Observatory for Refinement and Geometric Exploration

Project Proposal & Statement of Work

Nathan Herling, Project Manager

Team: Nathan Herling (solo capstone)

Faculty Advisor: Professor Ken Johns (UA ATLAS HEP Group)

Date: June 5, 2026

Project documentation site:
n-herling-mk1.github.io/INFO_698_documentation

This is a ***hybrid proposal*** *combining the Software and Research project templates: Sections 2 and 3 each carry both a product/software track and a research track (the 2b and 3b subsections).*

Revision History

Version	Summary of Changes	Date
0.1	Converted hybrid (Software + Research) template to LaTeX; title page and Executive Summary drafted	06/01/26
0.2	Executive Summary reworked to the three-experiment elevator pitch; Sections 2/2b (user/market + literature), 3/3b (features + research deliverables), and 4 (Summer 2026 milestones + embedded Gantt) filled	06/05/26

Keep this as a high-level summary of major revisions only. For a short document like the SOW you may only need 2–3 versions.

Contents

The table of contents above is generated automatically. Use a page break before the Executive Summary, then start a new page where it makes sense for clarity, balanced with keeping excessive white space out of your document.

1. Executive Summary

This section was written by Nathan Herling.

FORGE (Functional-posterior Observatory for Refinement and Geometric Exploration) is a web-hosted observatory built around the question every trained model hides: *is its confidence earned, or has it simply run out of data?* The *eventual* product is a general platform where a researcher loads their own working model, flips it from a single point estimate to a full Bayesian posterior, and steers that uncertainty in real time — a closed-form Last-Layer Laplace Approximation (LLLA) for instant feedback, Hamiltonian Monte Carlo (HMC) for ground truth — serving two needs from one interface: exploring whether a model has reached its *epistemic limit*, and visualizing and tuning that uncertainty with Bayesian tools, with no pipeline for the user to build.

The `mk_1` release reported here is the *proof of concept* for that vision, not the general platform itself. Rather than accept arbitrary user-supplied models, `mk_1` reproduces three known, published results identified in the project's literature search — reproduced and preloaded by the author with documented performance — and demonstrates the Bayesian-exploration layer on those fixed reproductions. The science and the interface are real and end-to-end; what `mk_1` deliberately scopes out is open-ended model ingestion, which a later release would add.

The motivation is a gap in everyday practice. A conventional network returns one number per input and reveals nothing about how much an equally valid version of the same model — a different seed, bootstrap, or architecture — would disagree. Bayesian inference replaces the single weight vector with a posterior $p(w \mid D)$, turning every prediction into a distribution whose width is the model's *epistemic* uncertainty made visible (as distinct from *aleatoric*, irreducible noise in the data).[1] FORGE exposes that distinction directly in the browser and pairs it against the original deterministic baseline, lowering the barrier to interpreting model uncertainty for both specialists and reviewers. Because a posterior is computationally expensive, FORGE keeps most of the network deterministic and makes only the final layer Bayesian: the LLLA path is fast enough to resample within an interactive session, while HMC supplies the trusted reference.

The three reproductions span deliberately heterogeneous domains, to show the FORGE pattern generalizes across model classes: (1) music-genre recognition with a convolutional network and a Bayesian classification head (GTZAN); (2) phonon density-of-states regression with an $E(3)$-equivariant graph network (Chen et al., 2021); and (3) the ATLAS Run 3 search for long-lived hidden-sector scalars via displaced muon-spectrometer vertices. Reproducing three rather than one is also a redundancy strategy: if one proves intractable on the term timeline, the proof of concept still stands on the remaining two.

The work is carried out by Nathan Herling under Professor Ken Johns in the University of Arizona ATLAS High-Energy-Physics group, across three repositories (documentation, experiments, and software) and validated in continuous integration against a tiered set of calibration datasets, C0–C3.[2] By the end of the term, FORGE will deliver a deployed, web-hosted observatory that loads the author's preloaded reproductions, serves

[1] The aleatoric/epistemic distinction follows A. Kendall and Y. Gal, "What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision?", NeurIPS 2017.

[2] **C0–C3** are the project's *calibration tiers*: small analytic datasets of increasing difficulty whose ground-truth posteriors are known in closed form, so the platform's samplers and visualizations can be scored against them automatically on every push (continuous integration, CI). They are distinct from the benchmarked tier D4 (the FAIR Universe HiggsML dataset, NeurIPS 2024), which anchors the work to a public physics benchmark.

their HMC-sampled posteriors, and supports interactive geometric exploration over them. Table 1 lists the preliminary division of subsystem responsibilities.

Team Member	Subsystem responsibility
Nathan Herling	Posterior sampling pipeline (HMC) and data tiers (C0–C3)
Nathan Herling	Geometry & visualization engine (credible regions, function-space spread)
Nathan Herling	Web dashboard, hosting, and deployment
Nathan Herling	Validation against the ATLAS Run 3 long-lived-particle search

Table 1: *Preliminary Subsystem Responsibilities*

2. User/Market research

This section was written by Nathan Herling.

Overall market

FORGE sits in the uncertainty-quantification (UQ) tooling space for machine learning, which has grown alongside the adoption of Bayesian methods in the data-rich physical sciences (high-energy physics, condensed matter, materials). The existing landscape splits cleanly into five segments: deterministic ML “playgrounds” (no uncertainty), post-hoc posterior diagnostics shipped as libraries, probabilistic-programming frameworks, production UQ/monitoring dashboards, and interactive sampler explainers built on toy targets. None of these is a hosted, interactive instrument for asking whether a *trained* model’s confidence is earned. That gap — between a code-heavy probabilistic framework and a static decision plot — is FORGE’s opening.

Target users

FORGE answers two questions with one instrument, so it serves two user types. They are not redundant: the first asks a decision question with a cost attached; the second asks an open “how” question. The diagnostician is the motivating problem (what makes the work worth doing); the observatory is the capability problem (what the tool literally does, and what hypotheses H1–H3 test). Each is framed below as User + Need + Because, with the design-actionable insight in the “Because” clause.

User 1 — Diagnostician	*"Should I collect more data, or is this as good as it gets?"*
User	Graduate researchers in data-rich physical sciences who train their own neural-net classifiers/regressors but are *not* Bayesian-ML specialists.
Need	To see whether a trained model's confidence is earned or borrowed — whether its remaining error is epistemic (fixable with more/better data or a different architecture) or irreducible.
Because	A single point estimate hides what the model does not know, and standing up a full Bayesian pipeline from scratch is enough engineering that researchers skip it and ship overconfident models. *This is exactly why FORGE preloads working baselines.*

Table 2: *User 1 point-of-view (the motivating, diagnostic use)*

User 2 — Observatory	*"What is driving this uncertainty, and what happens if I change it?"*
User	Practitioners and students of Bayesian deep learning who want to understand how their modeling choices shape the posterior.
Need	An interactive instrument to reshape the posterior in real time — turn a prior, sampler setting, likelihood, or architecture knob — and immediately see the geometry respond.
Because	The normal Bayesian loop is slow (minutes per HMC run) and its outputs are scalars and trace plots, not geometry you can steer. *This is exactly why the closed-form LLLA fast path is non-negotiable for real-time knob response (hypothesis H3).*

Table 3: *User 2 point-of-view (the general, exploratory capability)*

Existing competitors

Each comparable below overlaps FORGE on at least one axis (interactive model exploration, posterior/uncertainty visualization, sampler diagnostics, or Bayesian experimentation), paired with the line that separates it from FORGE.

Comparable	How FORGE differs
MCMC Interactive Gallery; GP-regression demo (interactive sampler/Bayesian explainers) chi-feng.github.io/mcmc-demo tmpl.fi/gp	Animate samplers on *fixed* 2-D targets or 1-D GP regression; FORGE samples a *trained model's* weight posterior across real benchmarks and adds a deterministic-vs-Bayesian comparison.
TensorFlow Playground; ConvNetJS (deterministic NN sandboxes) playground.tensorflow.org cs.stanford.edu/people/karpathy/convnetjs	Decision-boundary sandboxes with *no* uncertainty; FORGE adds the epistemic layer they deliberately omit.
ArviZ; bayesplot; ShinyStan (post-hoc posterior diagnostics) python.arviz.org mc-stan.org/bayesplot mc-stan.org/shinystan	Plot samples you *already* have; FORGE is hosted and generates and reshapes posteriors in-session.
PyMC; Pyro; JASP (probabilistic-programming / stats) pymc.io pyro.ai jasp-stats.org	Code frameworks or a stats package; FORGE is the no-code "turn a knob, watch the posterior move" layer on top.
Weights & Biases; WhyLabs; Evidently AI (production UQ/monitoring) wandb.ai whylabs.ai evidentlyai.com	Track deployed-model metrics and drift; report confidence numbers, not posteriors or Bayesian reasoning.
Netron (architecture viewer) netron.app	Shows structure only — no uncertainty, posterior inference, or Bayesian reasoning.

Table 4: *Comparable tools and how FORGE is differentiated*

User insights and pain points

Two pain points recur across both user types and drive the architecture directly: (i) the engineering cost of standing up a Bayesian pipeline (HMC/NUTS, calibration, last-layer Laplace) is high enough that researchers skip it — addressed by preloading working baselines that users inherit rather than rebuild; and (ii) the edit-code-and-wait loop is slow and its outputs are scalars and trace plots rather than steerable geometry — addressed by the closed-form LLLA fast path that keeps the knob panel responsive, with heavy HMC reserved for ground truth. The problem framing and the technical design are the same claim seen from two sides.

2b. Literature Review/Market research

This section was written by Nathan Herling.

Because FORGE has both a software and a research element, novelty is gauged on two fronts: the comparable-tools scan above (so the build is clear of others' ground) and the "research of research" below — peer-reviewed work that grounds FORGE's ingredients and sharpens where its contribution is genuinely new. A continually-updated, browsable version of this review is maintained as the Literature Search panel on the project documentation site.

Methodological foundations

Each FORGE ingredient rests on established work. Bayesian deep-learning inference: variational Bayes-by-backprop (Blundell et al., 2015) and Monte-Carlo dropout as approximate inference (Gal & Ghahramani, 2016). The interactive fast path: the Last-Layer Laplace Approximation and its "Bayesian last layer" framing (Kristiadi et al., 2020; Daxberger et al., 2021, *laplace-torch*). The ground-truth path: Hamiltonian Monte Carlo and the No-U-Turn Sampler (Neal, 2011; Hoffman & Gelman, 2014). The aleatoric/epistemic decomposition that motivates the whole tool: Kendall & Gal (2017). The reproduced experiments draw on domain baselines: an $E(3)$-equivariant graph network for phonon DOS (Chen et al., 2021) over the DFPT phonon database (Petretto et al., 2018), and CNN-based genre recognition (Sridhar, 2024; and the author's prior BEARDOWN work, Herling & Attili, 2025).

Closest prior art and the novelty gap

The table below collects the work nearest to FORGE's *purpose*—a posterior over models used to decide where knowledge is lacking—and its *interactivity*.

Reference	Relevance to FORGE
Dearden, Friedman & Andre — *Model-Based Bayesian Exploration* (UAI 1999)	An early statement of the core FORGE logic — a posterior over models beats a point estimate for deciding where knowledge is lacking — but applied as an RL exploration algorithm, not a visualization tool.
Frazier — *A Tutorial on Bayesian Optimization* (2018)	Shares the "visualize posterior uncertainty, decide where you're unsure" pattern, but aimed at optimizing a function rather than diagnosing a trained model.
Wang, Jin, Schmitt & Olhofer — *Recent Advances in Bayesian Optimization* (ACM CSUR 2023)	Confirms the BO field is optimization-centric; useful for positioning FORGE's model-*diagnosis* purpose as distinct.
Gabry, Simpson, Vehtari, Betancourt & Gelman — *Visualization in Bayesian Workflow* (JRSS-A 2019)	The methodological backbone for FORGE's plots — but static, R-package workflow guidance, not an interactive hosted dashboard.
Nussbaumer, Böck & Cito — *Online and Interactive Bayesian Inference Debugging* (ICSE 2026; *InferLog Holmes*)	The closest prior art to FORGE's online, interactive ambition — but an IDE debugger for PPL code aimed at *fixing inference*, not a hosted multi-domain dashboard for exploring a trained model's epistemic uncertainty.

Table 5: *Research landscape — closest prior art*

Where FORGE is novel

The landscape grounds each ingredient individually, but no existing system is a hosted dashboard that simultaneously (1) preloads trained models across heterogeneous domains, (2) toggles deterministic $\leftrightarrow$ Bayesian and compares posteriors against the baseline, and (3) reshapes posterior geometry in real time through a knob panel backed by closed-form LLLA plus heavy HMC. The combination, framed by the question *"is this model's confidence earned or borrowed—has it reached its epistemic limit?"*, is the differentiated contribution.

3. Product Features

This section was written by Nathan Herling.

FORGE is built in three maturity rungs of *one* product, not three different products: the same application (data locked, methodology exposed) is carried up a ladder — first proven locally (*MVP — phase 1*), then deployed (*MVP — deployed*), then optionally opened to an LLM agent (*Stretch — MCP*). The science underneath all three is identical, so the project always has a complete deliverable no matter how high the ladder is climbed. Phase 1 (features F1–F5) is everything required to demonstrate the concept locally; the deployed rung (F6) ships the same stack to a cloud host; only the MCP rung (S1) is a true stretch, and it adds no new science. All features are owned by Nathan Herling (solo capstone).

#	Feature	What it does / why the user wants it
MVP — phase 1 (minimum feature set; local + logging)		
F1	Preloaded multi-domain baselines	Three working, documented models (genre, phonon, ATLAS) load instantly; the user inherits a baseline instead of building a pipeline.
F2	Deterministic $\leftrightarrow$ Bayesian toggle	Flip a trained model from a point estimate to a posterior and compare the two side by side — the core "is the confidence earned?" view.
F3	Real-time knob panel	Turn prior, sampler, likelihood, and architecture knobs and watch the posterior geometry respond live, backed by the closed-form LLLA fast path.
F4	Posterior-geometry visualization	Credible regions, function-space spread, and the per-config posterior-σ distribution, rendered in-browser.
F5	Resource-logging layer	Per-run telemetry (memory, time, cost) that turns Phase-2 infrastructure sizing into an empirical fit rather than a guess.
MVP — deployed (same stack, cloud host)		
F6	Cloud deployment	The same Docker images run on a cloud host; only one environment variable changes between local and deployed.
Stretch — MCP (optional agent interface)		
S1	MCP server	A thin manifest advertises the compute functions as tools an LLM client (e.g. Claude Desktop) can call — no new infrastructure.

Table 6: *FORGE feature list across the three MVP rungs, all owned by N. Herling*

Feature F3: Real-time knob panel — the load-bearing interactive feature

The whole value of FORGE for User 2 depends on the knob panel feeling instant, which is why the architecture splits into two compute paths: a closed-form LLLA path on an always-on CPU for interactivity, and a heavy HMC/NUTS path on a serverless GPU for ground truth. The performance envelope is summarized in Table 7.

Parameter	Target	Ceiling	Comments
LLLA knob-to-update latency	$\leq 1\,\mathrm{s}^{*}$	$\leq 3\,\mathrm{s}^{*}$	Closed-form last-layer; must feel interactive
HMC "ground-truth" resample	$\sim$1 min*	$\leq$5 min*	Serverless GPU, scale-to-zero; per-job, not interactive
Compute paths exposed	2	2	LLLA (CPU, always-on) and HMC/NUTS (GPU, on demand)

* *Early planning estimates, not yet profiled — to be revisited once Phase 1 and its resource-logging layer (F5) are complete and the fitted time-vs-scale curves* ($\mathrm{time} \approx f(n_{\mathrm{params}}, n_{\mathrm{samples}}, \ldots)$), *including data size) are available. (Cold-start container spin-up and JAX/XLA compilation on the serverless GPU are not yet included in the HMC figures.)*

Table 7: *Feature F3 interactive-path parameters*

Feature F6: Cloud deployment (MVP — deployed) — hosting envelope

Deployment must hold to a hobby-scale budget; the only component that must run continuously is one always-on CPU box, with the heavy GPU path serverless and effectively $0 at idle. Planning estimates (as of May 2026) are in Table 8.

Layer	Est. cost/mo	Notes
Frontend (static, GitHub Pages)	∼$1	Domain only; Pages + TLS free
Backend API + LLLA (Fly.io/Render)	$10–25	Always-on CPU box, ∼2 GB RAM
HMC heavy jobs (Modal, serverless GPU)	∼$0 baseline	Scale-to-zero, per-second; credits cover dev
Database	$0	Stateless / explicit job-ID design; deferred
Target total	**∼$25**	Size of one always-on CPU box

Table 8: *Feature F6 hosting-cost envelope*

3b. Research Project Deliverables

This section was written by Nathan Herling.

Final presentation format

The project culminates in (i) a written report with introduction, methods, and results sections and at least five peer-reviewed citations, carrying at least two informative graphics (the deterministic-vs-Bayesian comparison and the fitted resource-cost curves); (ii) an iShowcase poster and live demo; and (iii) the deployed FORGE site plus three public GitHub repositories (documentation, experiments, software), tagged `v1.0` at freeze.

What analysis is being run

Three reproductions, each driven to a functional ground-truth result before the next begins, then extended with Bayesian inference:

- **Genre recognition** — a CNN with a Bayesian classification head over GTZAN (spectrogram + engineered-feature encoders).
- **Phonon DOS** — an $E(3)$-equivariant graph network with a Bayesian last-layer head over the 51-point DOS output; the published baseline gives point predictions only, so calibrated credible intervals are FORGE's added contribution.
- **ATLAS LLP** — a deterministic MLP discriminating signal (HSS $\to$ LLP) from QCD background, extended with HMC over the output layer.

Across all three, the analysis is posterior inference via the closed-form LLLA fast path and ground-truth HMC/NUTS, evaluated for calibration coverage in addition to raw accuracy.

What accuracy is expected

Each reproduction commits up front to targets drawn from its source work — FORGE's "empirical contract" for declaring a baseline reproduced. Calibration coverage is the additional contribution layered on top.

Experiment	Reproduction target
Genre recognition	Best-config validation accuracy ≥ 0.75 (cfg_14: 0.766); sweep mean ≈ 0.72; per-genre F1 in 0.46–0.89 (BEARDOWN, Herling & Attili 2025).
Phonon DOS	$\geq 70\%$ of test materials within 10% relative error on the average phonon frequency; per-element DOS MSE 10^{-3}–10^{-1} (Chen et al. 2021).
ATLAS LLP	**#TODO** — reproduction target not yet decided; to be set within the next week (with advisor input).

Table 9: *Per-experiment reproduction targets*

What if the analysis doesn't work

A null result is acceptable and, in this project, often *informative*: a model shown to sit at its epistemic limit is exactly the diagnostic FORGE exists to surface. Risk is further managed by redundancy — three reproductions mean the platform still demonstrates the concept if one proves intractable. The descope order is fixed: drop the MCP stretch (S1) first, then trim the depth of the Bayesian/visualization refinements; the three baseline reproductions and the testing weeks are never cut, because they carry the write-up.

What if the data isn't available

There is no scraping dependency. All three datasets are public or already in hand: GTZAN (genre); the DFPT phonon database via the Materials Project (phonon, Petretto et al. 2018); and the ATLAS reproduction inputs from the author's prior thesis work and Monte-Carlo samples. The C0–C3 calibration tiers are analytic, so their ground truth is always available for the CI scoring path regardless of external data.

4. Project Timeline & Gantt Chart

This section was written by Nathan Herling.

The thirteen-week schedule (single contributor) is anchored at Week 0 = May 18, 2026. The substantive work — standing up the three reproductions inside the Phase-1 architecture — is front-loaded so that all three are integrated by Week 5 (the guaranteed "floor" deliverable); deployment, the Bayesian/visualization layer, and the MCP stretch follow, with a dedicated testing/buffer block before the paper and hand-in. Table 10 lists the dated milestones; Figure 1 is the corresponding Gantt.

Milestone	Date
Team formation; three repositories bootstrapped (course)	5/22/26
Signed proposal v1 (course)	5/29/26
Phase-1 architecture scaffolded	5/29/26
Genre-recognition reproduction functional (ground truth)	6/5/26
Phonon DOS reproduction functional (ground truth)	6/12/26
ATLAS LLP reproduction functional (ground truth)	6/19/26
Three experiments integrated into Phase-1 stack (floor)	6/26/26
Stack deployed — MVP (same images, cloud host)	7/3/26
Bayesian + visualization layer across all three	7/10/26
MCP server (stretch) tested via Claude Desktop	7/24/26
Testing complete — pipelines + platform end-to-end	7/31/26
Final paper drafted; iShowcase poster + demo prep	8/7/26
Final submission & iShowcase; repo freeze + v1.0 tag (course)	8/12/26

Table 10: *Milestone Schedule (Summer 2026)*

The Gantt chart in Figure 1 mirrors this schedule. Its source of truth is two hand-edited files in the documentation repository: `data/gantt.json` drives the bars (per-task phase, start/end week, and status), while `data/weekly_todos.json` holds the granular per-week task lists — each a `{name, status}` entry keyed to weeks W0–W12 — that populate the interactive Gantt's clickable week-detail panel on the project site. A live version (clickable per-week detail, downloadable PNG) is maintained at the documentation site linked below.

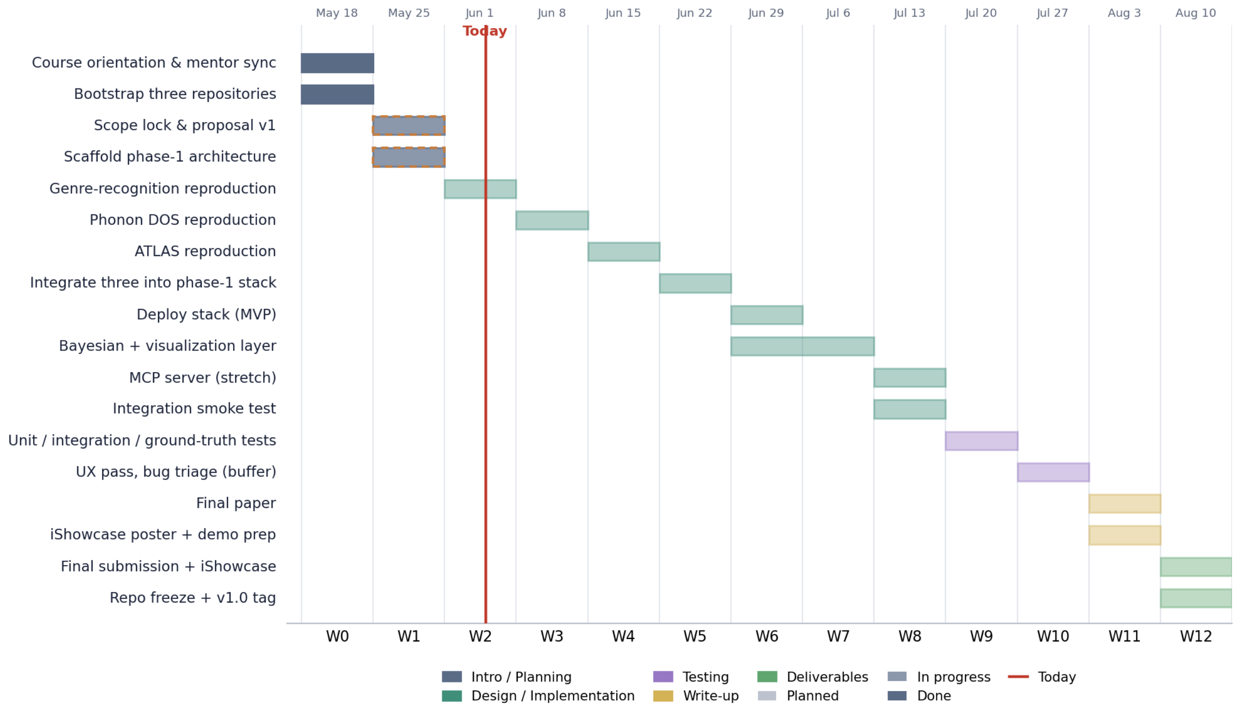

Figure 1: *FORGE project Gantt (snapshot). Bars span the weeks each task is active; fill density encodes status (planned / in progress / done); the vertical line marks "today". Live version: n-herling-mk1.github.io/INFO_698_documentation #gantt.*

5. Ethics

Data ethics is a key consideration in any product we create. Include a completed ethics chart (see examples on D2L). For each question, answer Y / N / M (Yes / No / Maybe) for general use and for the data-breach scenario.

#TODO

ATLAS data ethics. The generic chart below does not yet address the ATLAS/CERN-specific data-handling obligations: collaboration data-access and publication policy, restricted-data and embargo handling, and any required internal review/approval before quoting or releasing results. Add an ATLAS-specific data-ethics treatment here (with advisor input) before submission.

#	Question	Generally	Data Breach
1	Could a user sell drugs or other illegal items on your platform?	Y/N/M	Y/N/M
2	Could a user of your platform engage in sex trafficking?	Y/N/M	Y/N/M
3	Could a user sell class notes or cheat on their homework on your platform?	Y/N/M	Y/N/M

#	Question	Generally	Data Breach
4	Could a stalker use your project to find someone?	Y/N/M	Y/N/M
5	Could your app be used to spy on or track individuals?	Y/N/M	Y/N/M
6	Could your app/software access the camera or microphone and record things without users being aware?	Y/N/M	Y/N/M
7	If someone uses your platform, could they be re-traumatized or have their mental health impacted in some way?	Y/N/M	Y/N/M
8	Could your algorithm promote material that would traumatize or upset individuals?	Y/N/M	Y/N/M
9	Would your users be upset if the data you collect was given to someone else?	Y/N/M	Y/N/M
10	Could a data leak potentially lead to identity theft?	Y/N/M	Y/N/M
11	If your site was hacked, would users potentially lose their job, spouse, or family?	Y/N/M	Y/N/M
12	Should there be an age limitation on your product?	Y/N/M	Y/N/M
13	Could someone use your product to find, contact, and potentially commit elder abuse?	Y/N/M	Y/N/M
14	If the data on your platform was breached, could it be used to blackmail the users?	Y/N/M	Y/N/M
15	Does the existence of your project imply that a particular racial group, gender, religion, or other protected category is inherently bad, gross, or unwanted?	Y/N/M	Y/N/M
16	Could your product be used to commit hate crimes against a specific group?	Y/N/M	Y/N/M
17	Does the primary content of your game or algorithm focus on something considered deeply unethical?	Y/N/M	Y/N/M
18	Does your game or software contain race, gender, or other stereotypes?	Y/N/M	Y/N/M
19	Could users of your app scam other individuals?	Y/N/M	Y/N/M
20	Is your particular algorithm biased toward predicting correctly only for one race, gender, or other group?	Y/N/M	Y/N/M
21	Are the users of your project aware of how their data will be used?	Y/N/M	Y/N/M
22	What are the possible misinterpretations of your results?	Y/N/M	Y/N/M
23	Does the use or purchase of your data potentially contribute to a dangerous group or regime?	Y/N/M	Y/N/M
24	Could your virtual reality environment cause injury to the user?	Y/N/M	Y/N/M

#	Question	Generally	Data Breach
25	Are your study participants or game players aware that their data will be collected and used?	Y/N/M	Y/N/M
26	Does your game or app contain addictive design elements without benefit to the user?	Y/N/M	Y/N/M
27	Does your survey contain an aspect of compulsion or unusually large incentive that would command users to take it even to their detriment?	Y/N/M	Y/N/M
28	Could your research outcomes harm an individual or entity?	Y/N/M	Y/N/M

Table 11: *Data Ethics Chart*

6. Approvals

The project proposal needs the approval of everyone to move forward. If approval is denied, rework the proposal and begin the approval cycle again. Consult the faculty advisor often. After the faculty advisor and team are comfortable with the SOW, only then should it go to the sponsor.

Type the names in the "Approver Name" column; written signatures only in the "Signature" and "Date" columns. Give advisors and sponsors at least 3 business days of lead time. Your instructor signs after grading, if the document passes. Keep the language below.

The signatures of the people below indicate an understanding of the purpose and content of this document by those signing it. By signing this document, you indicate that you approve of the proposed project outlined in this Statement of Work, the division of work, the Ground Rules, and that the next steps may be taken to create a Product Specification and proceed with the project.

This document is based upon and supersedes the <PRD title> Version X.X. Deviations (versus clarifications) from the PRD have been clearly noted. For any requirements not listed in this SOW, the PRD requirements shall remain in effect.

Approver Name	Title	Signature	Date
Type names here	Team Project Manager		
All team members must sign	Team Member		
	Team Member		
	Advisor		
	Instructor		

Before you submit to advisor/sponsor for review: refresh the table of contents; verify every table and figure has a number and caption with at least one introductory sentence; clean up white space and page breaks; make sure the revision block is correct.

Before you submit for grading, update the author table below. It need not be 100% accurate but should give the instructor a good idea of how much each person contributed. For tables and diagrams just put n/a for the word count.

Section	Author	Word Count
1. Executive Summary	Nathan Herling	tbd
2 / 2b. User/Market & Literature Review	Nathan Herling	tbd
3 / 3b. Product Features & Research Deliverables	Nathan Herling	tbd
4. Timeline & Gantt (incl. tables/figure)	Nathan Herling	n/a

7. Appendix

A. Advisor Engagement

Project Team Responsibilities

- The Project Manager will set up and facilitate a weekly call/meeting with the Faculty Advisor. The Project Team will provide weekly status updates including upcoming deliverables, critical issues, and any adjustments to the Project Plan.
- Documents will be provided to the Faculty Advisor with adequate time for review and signature, with a minimum review time of 3 days prior to the document due date.
- Design files will be provided to the Faculty Advisor as requested in an agreed format.
- Support requirements will be clearly requested with the dates required and adequate time for fulfillment.
- Modification requests to the Project Plan by the Faculty Advisor will be reviewed and agreed to within 1 week of the request.

Faculty Advisor Responsibilities

- The Faculty Advisor will provide knowledge and expertise to help the group stretch their skills.
- The Faculty Advisor will participate in a weekly or bi-weekly call/meeting to review status, upcoming deliverables, priorities, issues, and progress to the agreed Project Plan.
- The Faculty Advisor will provide document review, feedback, and approval (or rejection, or approval with contingencies) with adequate time for the team to meet course due dates.
- The Faculty Advisor will provide feedback on support requirements, including guidance on design decisions, design files, test plans, test procedures, and test results.

- The Faculty Advisor shall provide technical advice and guidance, answering inquiries approximately 1 hour per week.
- Modifications to the Project Plan by the Project Team will be resolved and documented within 1 week of the request.
- Grade the finalized project using a skill-based rubric.
- Attend iShowcase in May.

B. Ground Rules

How the team will conduct the business of completing this project. Each team member must sign the Approvals section indicating their acknowledgement of these Ground Rules. Do not remove items from this list, but you may add items.

As a team and as individual team members, we agree to:

- **Stay focused on our objectives and goals.** Each time the team meets, we will clearly define our objectives and desired outcomes at the beginning of the meeting, and politely remind team members if we are getting off track.
- **"Sidebar" issues that are relevant but not consistent with the immediate objectives.** Create a "sidebar" where off-topic but important matters can be listed and discussed later.
- **Listen when others are speaking.** We will listen and consider others' input before adding our own comments.
- **All viewpoints will have an opportunity to be heard.** We will make an effort to get each team member's viewpoint and ensure no one dominates the discussion.
- **Differences of opinion will be discussed respectfully.** We will identify areas of agreement before assessing disagreement, weigh the merits of different opinions, and agree on a process for choosing a direction. All team members will respect and follow the decision.
- **Look for the good points in new ideas.** We will explore the value in each idea as we assess and select our path forward.
- **Focus on the future, not the past.** We will use past experience to inform decisions but focus the discussion on future objectives; blame is counterproductive.
- **Agree upon specific action items and next steps.** At the end of each meeting we will summarize and agree on specific next steps, action items, and assignments.
- **Accountability.** Each member is responsible for their individual assignments and contribution to the team objectives. We will honor our responsibilities and not let our team members down.